

3(a)	(work done =) force $\times$ displacement in direction of the force	B1
3(b)(i)	1. $(\Delta)E = mg(\Delta)h$	C1
	$= 0.42 \times 9.81 \times 78$ $= 320 \text{ J}$	A1
	2. $E = \frac{1}{2}mv^2$	C1
	$(\Delta)E = \frac{1}{2} \times 0.42 \times 23^2$ $= 110 \text{ J}$	A1
3(b)(ii)	work done = $320 - 110$ (= 210 N)	C1
	average resistive force = $210 / 78$ $= 2.7 \text{ N}$	A1
3(c)	downward sloping line from (0, g) to a non-zero value on the time axis	M1
	line is curved with a gradient that becomes less negative and the line meets t -axis at time $t < T$	A1